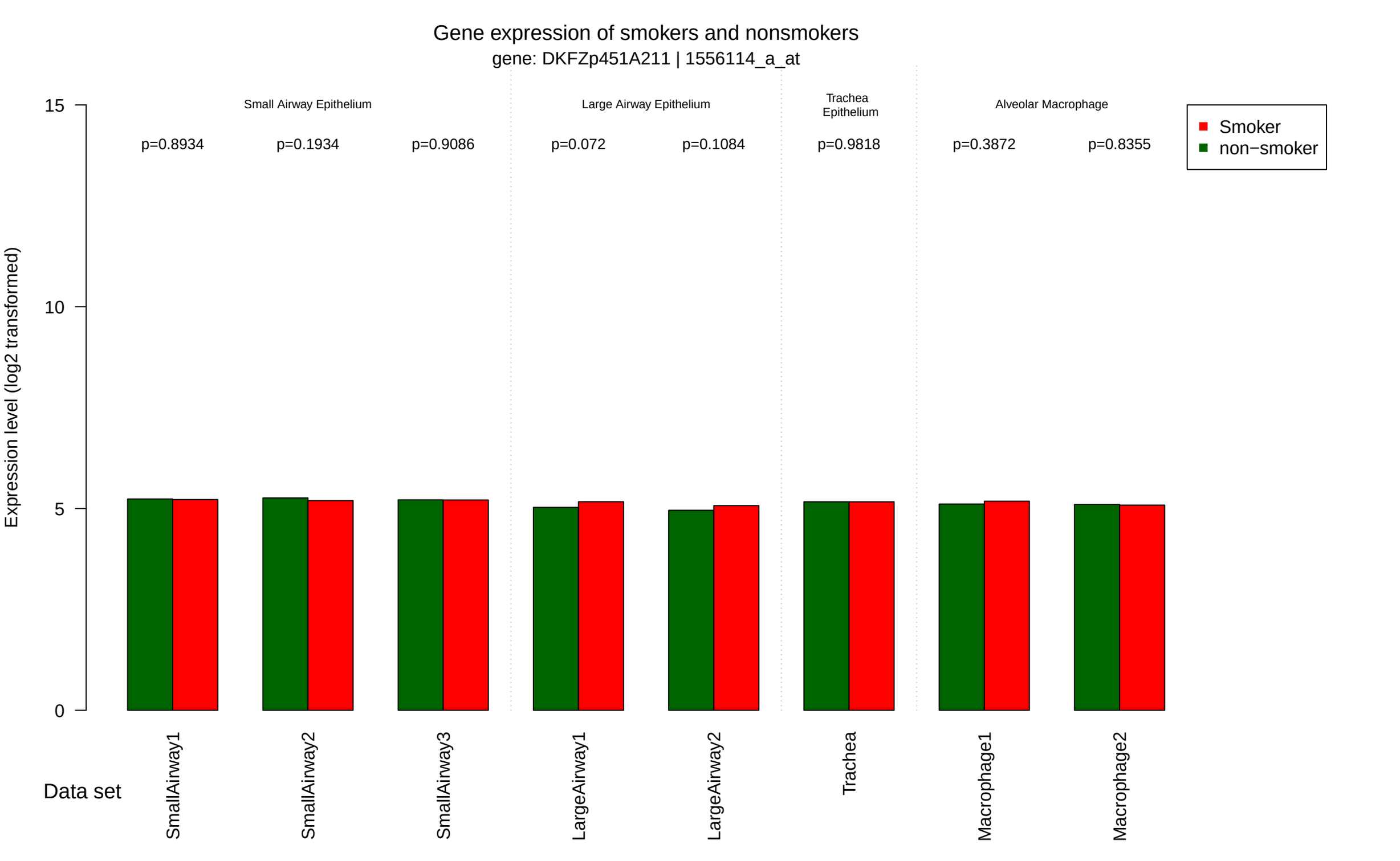
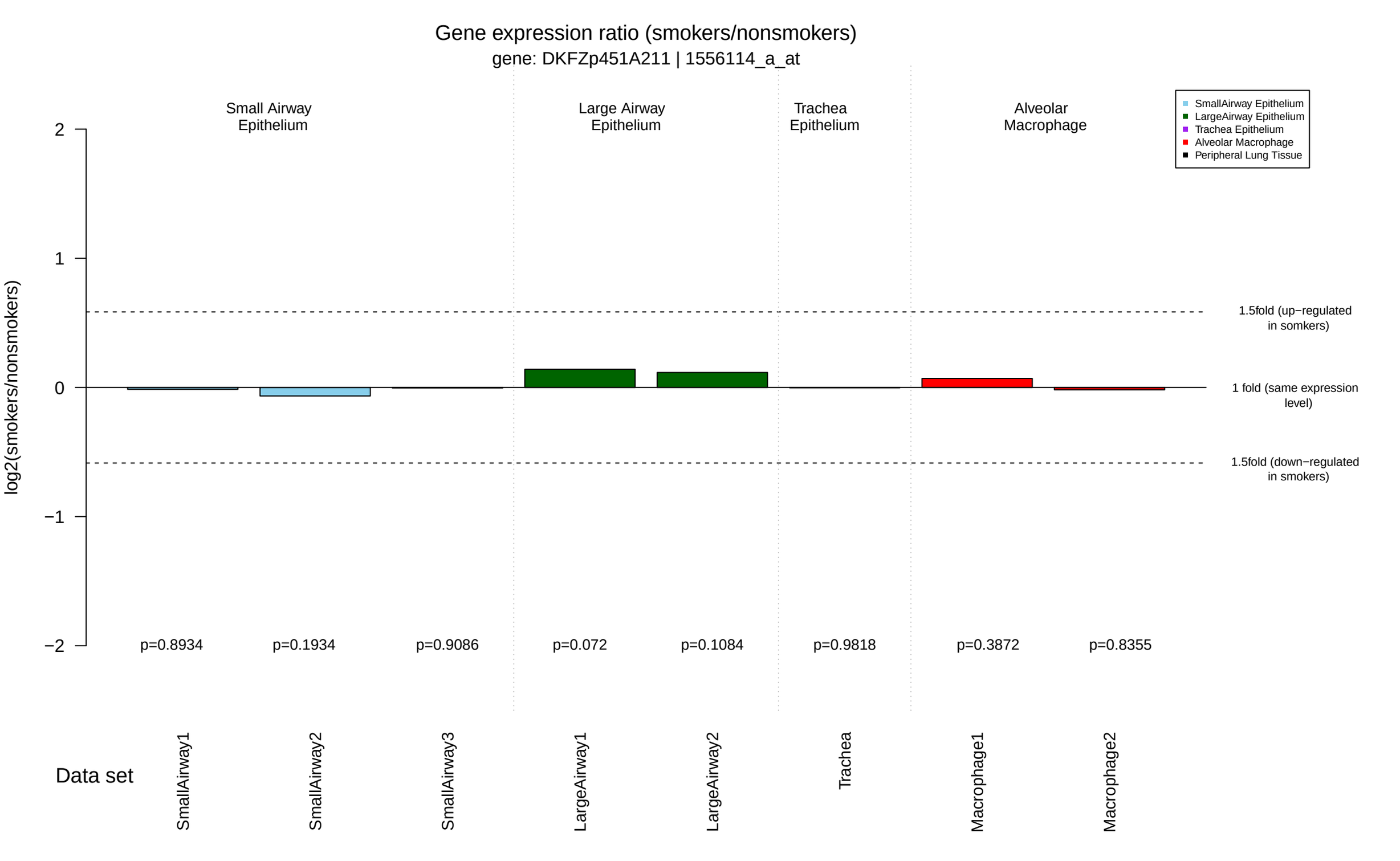
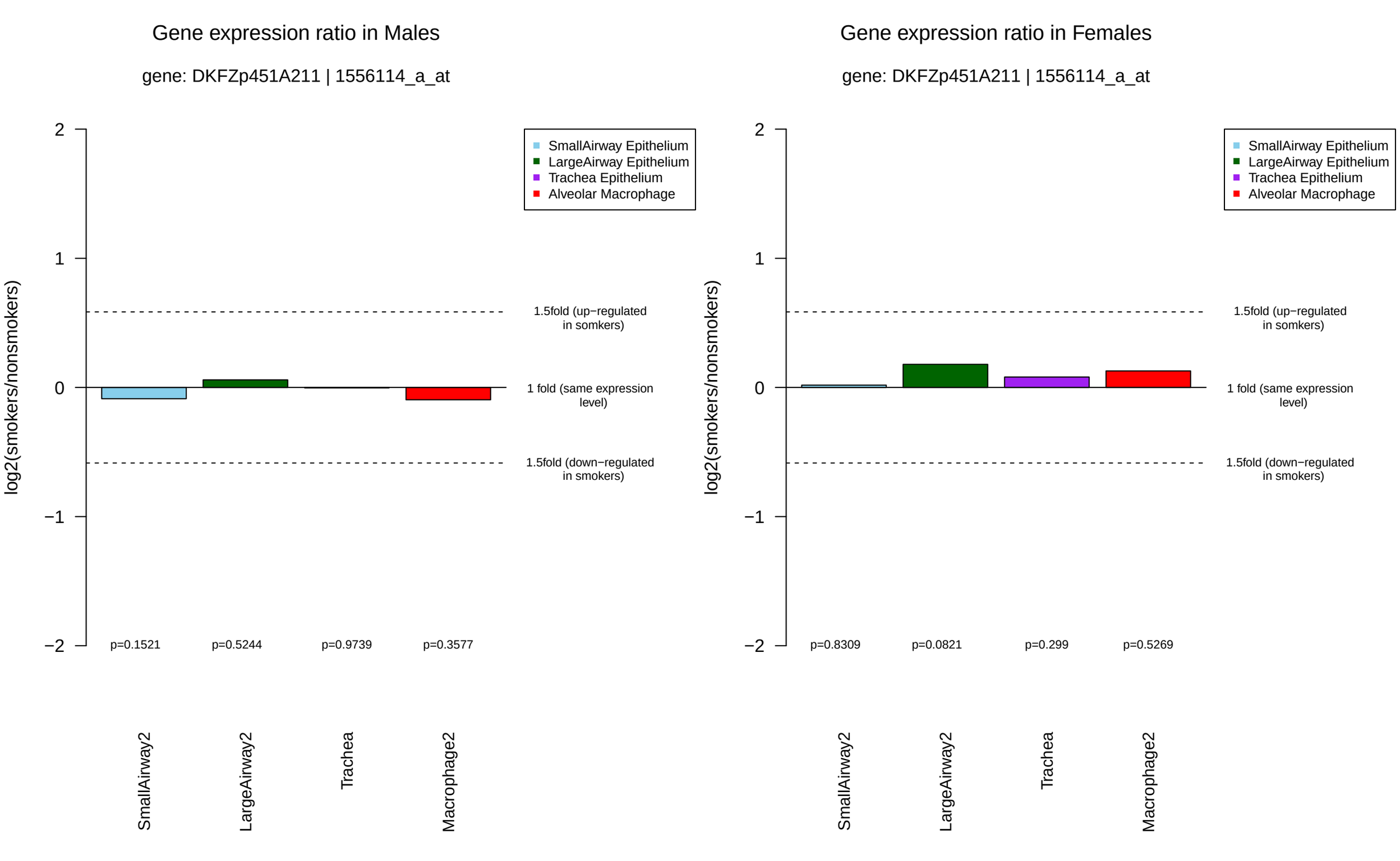
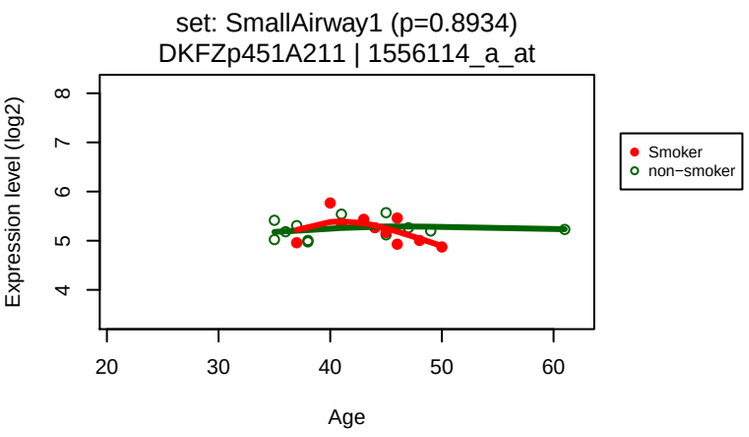
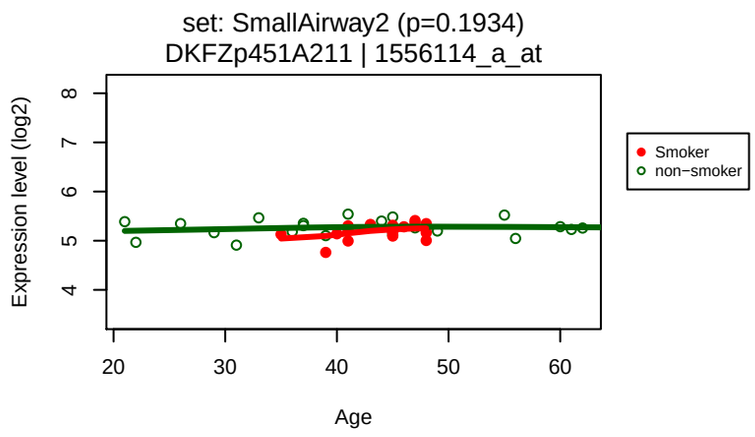
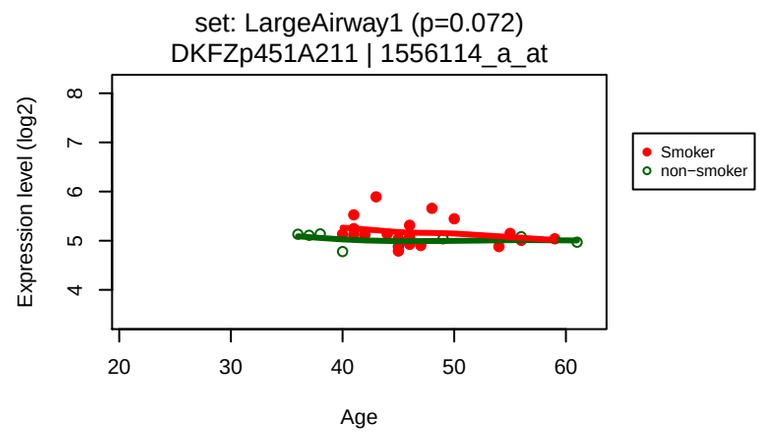

Gene expression of smokers and nonsmokers
gene: DKFZp451A211 | 1556114_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8934
p=0.1934
p=0.9086
p=0.072
p=0.1084
p=0.9818
p=0.3872
p=0.8355
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

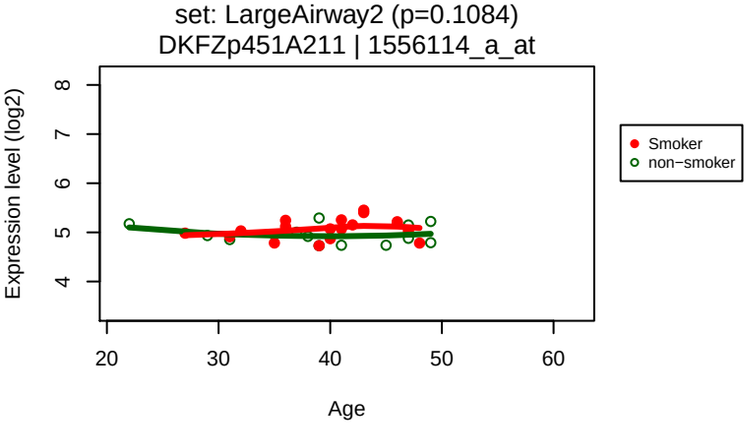

Gene expression ratio (smokers/nonsmokers)
gene: DKFZp451A211 | 1556114_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8934
p=0.1934
p=0.9086
p=0.072
p=0.1084
p=0.9818
p=0.3872
p=0.8355
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

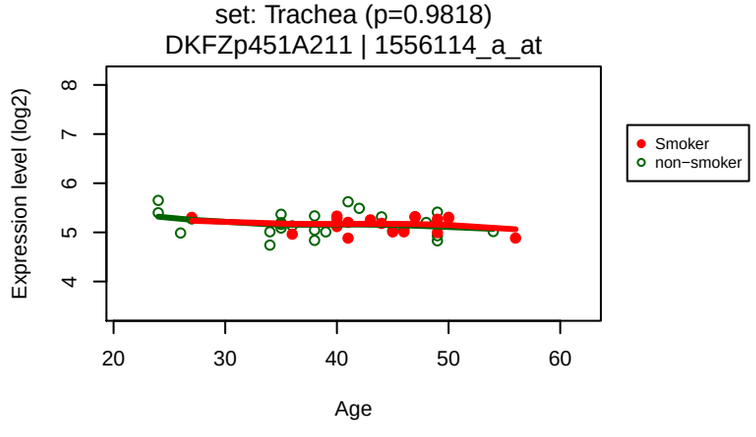

Gene expression ratio in Males
gene: DKFZp451A211 | 1556114_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1521
p=0.5244
p=0.9739
p=0.3577
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DKFZp451A211 | 1556114_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8309
p=0.0821
p=0.299
p=0.5269
SmallAirway2
LargeAirway2
Trachea
Macrophage2

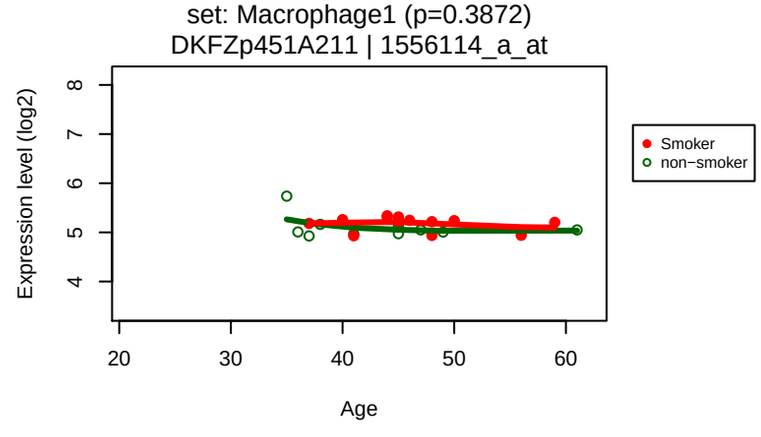

set: SmallAirway1 (p=0.8934)
DKFZp451A211 | 1556114_a_at
Expression level (log2)
Age
Smoker
non−smoker

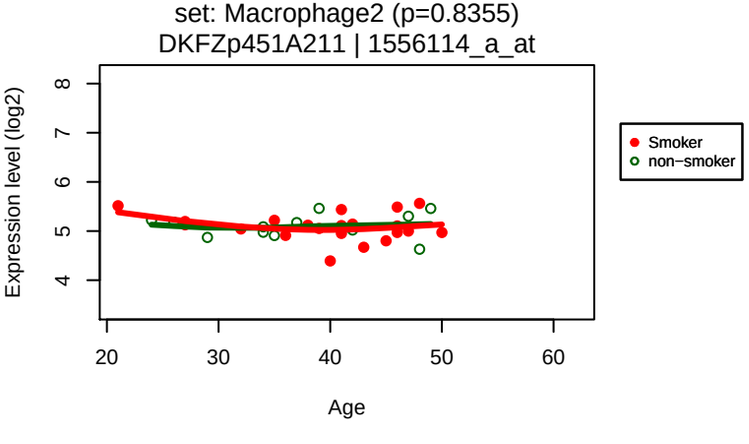

set: SmallAirway2 (p=0.1934)
DKFZp451A211 | 1556114_a_at
Expression level (log2)
Age
Smoker
non−smoker

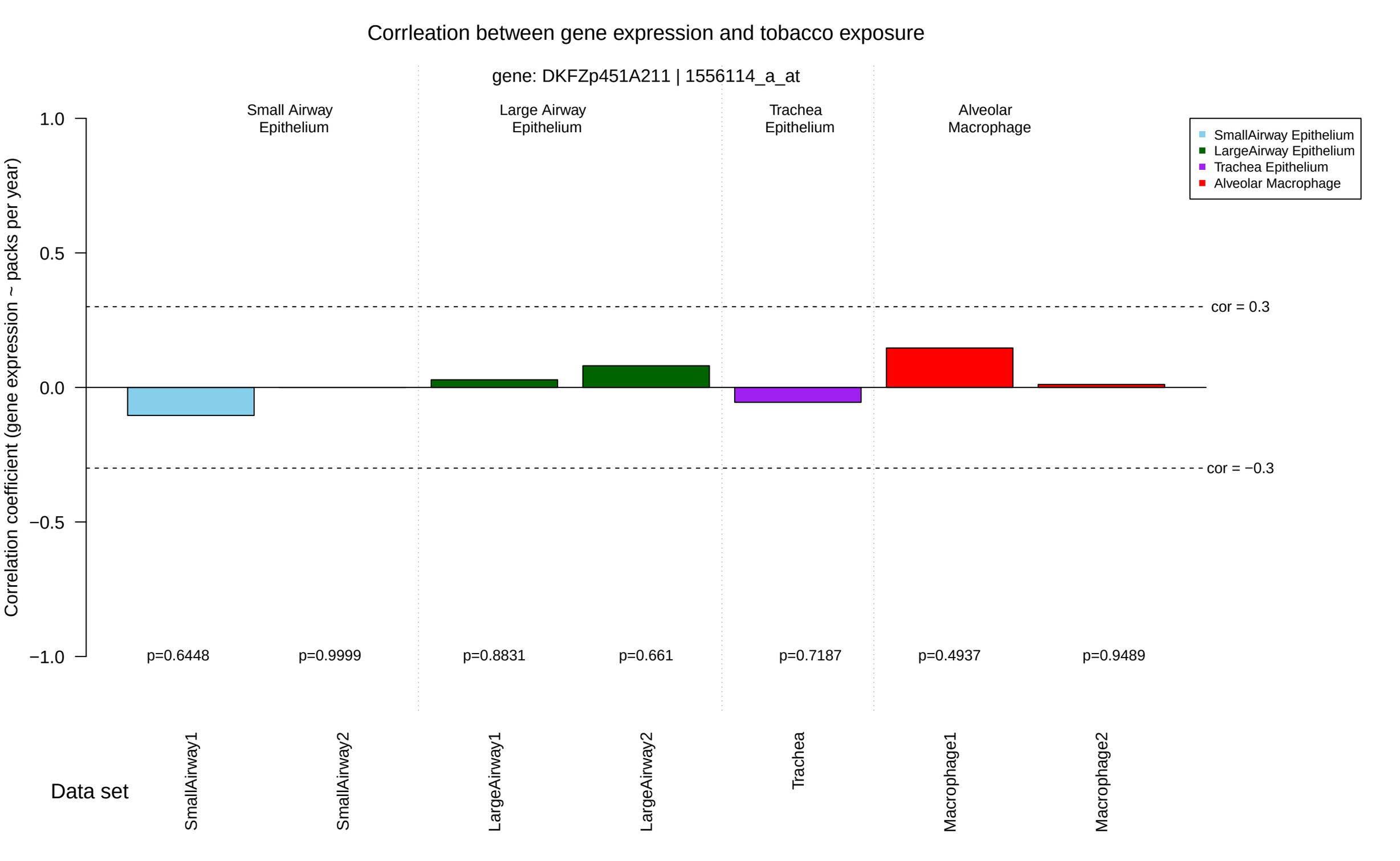

set: LargeAirway1 (p=0.072)
DKFZp451A211 | 1556114_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1084)
DKFZp451A211 | 1556114_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9818)
DKFZp451A211 | 1556114_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3872)
DKFZp451A211 | 1556114_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8355)
DKFZp451A211 | 1556114_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DKFZp451A211 | 1556114_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6448
p=0.9999
p=0.8831
p=0.661
p=0.7187
p=0.4937
p=0.9489
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2